Sayak Roy

AI Automation & Platform Engineer

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Professional Summary

AI Automation & Platform Engineer with 3+ years of experience across **network automation**, **cloud**, **DevOps**, and **edge operations**. At **DERQ**, I maintain **AI-powered Intelligent Transportation Systems** and built **Derq Ops**, an operations platform with **SSH health checks**, **Playwright automation**, and an **MCP-connected AI agent**. At **Tata Communications**, I automated **SDN/BGP provisioning** (**80%** less manual effort) and built **ML traffic forecasting**. I also ship **full-stack AI products** with **Next.js**, **Supabase**, and **Claude/OpenAI APIs**.

Skills

- **Cloud & Infrastructure – AWS** (EC2, Lambda, S3, CloudWatch, API Gateway, EKS, ECS), **Azure**, **Terraform**, **Ansible**, AWS CDK, CloudFormation, NGINX, **Docker**, **Kubernetes**.
- **Networking & SDN – OpenDaylight**, **NETCONF/RESTCONF**, **YANG/OpenConfig**, **BGP**, **OSPF**, **IS-IS**, **EIGRP**, **BMP**, **bgpq3**.
- **Edge & Ops – Linux**, **Docker**, **SSH/paramiko**, **RTSP/ffprobe**, **VPN**, **TCP/IP**, **OpenCV**.
- **Programming & Frameworks – Python**, **JavaScript**, **TypeScript**, HTML, MATLAB, **Next.js**, **Streamlit**, **FAST API**, **Flask**, **Selenium**, **Playwright**.
- **Operating Systems – Linux** (Ubuntu, Debian), Windows, MAC OS.
- **DevOps & CI/CD – GitHub Actions**, **Jenkins**, **GitLab CI/CD**, Cloud-native deployment, **pytest**.
- **Machine Learning & AI** – Time-Series Models (**ARIMA**, **SARIMAX**, **Prophet**, **XGBoost**, **LSTM**, **TimeGPT**), Ensemble Learning, Anomaly Detection (**Isolation Forest, LOF, One-Class SVM**), Clustering (**K-Means**, **DBSCAN**), **CNN**, **RNN**, **Claude/OpenAI APIs**.
- **Full-Stack Development – React.js**, **Next.js**, **Tailwind CSS**, Express.js, Flask, **RESTful API Design**, JWT Auth, Environment Management, **PostgreSQL**, **Supabase**, **MongoDB**, **SQL**.
- **Data Engineering & Automation – Pandas**, **NumPy**, **OpenPyXL**, **XlsxWriter**, **python-pptx**, **ETL Pipelines**, Excel/PPT Automation, MIS Reporting, API Integrations.
- **Monitoring & Visualization – Grafana**, **Prometheus**, Kentik, Power BI, **Chronograf**, **Telegraf**, **CloudWatch**.
- **Tools & Platforms – Git**, **GitHub**, Power Automate, SharePoint, **n8n**, **LangChain**, **LangGraph**, **MCP**, **ChromaDB**, **FAISS**, **Pinecone**, MS Forms.

Education

SRM University, B.Tech in Computer Science (Specialization: Cyber Security)

Experience

Deployment & Maintenance Software Engineer, DERQ – Dubai, UAE (Remote) December 2025 – Present

- Deploy, configure, and maintain AI-powered Intelligent Transportation Systems (ITS) across distributed edge environments using **Linux**, **Docker**, and **Python**.
- Troubleshoot production issues involving **Edge Processors**, **RTSP/IP cameras**, **VPN**, **TCP/IP**, **SSH**, and application services through log analysis and root cause identification.
- Develop Python automation tools for deployment, system validation, monitoring, and operational maintenance, improving deployment efficiency and reducing manual effort.
- Collaborate with software, AI, and infrastructure teams to perform remote deployments, software upgrades, configuration management, and production support across global customer sites.
- Built **Derq Ops**, a local **operations platform** (**Python 3.12**, **FastAPI**, **Jinja2/HTMX**, **SQLite**) consolidating daily site maintenance.
- Automated the daily **backlog report** (**Slack API**, **Playwright**-driven DERQ dashboard, **SSH**) with **Excel/CSV** export and **Slack** post.
- Implemented **Priority 1/2/3** SSH checks (**paramiko**), incl. **RTSP checks** with **ffprobe** and **one-click service restarts**.
- Built **camera-movement detection** using **OpenCV** feature matching against reference frames.
- Integrated an **AI agent** with **MCP connectors** (Gmail, Calendar, Drive, Slack, GitHub) and **in-browser voice**.
- Covered the platform with **230 automated tests** (**pytest**), including **end-to-end Playwright tests** against a mock dashboard.

AI Infrastructure & Automation Engineer, Tata Communications – Chennai, Tamil Nadu April 2023 – December 2025

- Automated router configurations using **OpenDaylight SDN Controller** with **NETCONF/RESTCONF APIs**, reducing manual provisioning effort.

- Designed YAML-based **OpenConfig templates** for multi-vendor routing protocols (BGP, OSPF, IS-IS, EIGRP).
- Built **RESTful APIs** to fetch live routing data and push configs, improving operational efficiency and reliability by almost 30%.
- Automated **IRR-driven Route Paths and ASN List/AS Set** using **bgpq3** and custom Python scripts to fetch, parse prefix lists and as-path sets from IRR sources, reduced manual effort by 80%.
- Built and deployed advanced forecasting systems using **ARIMA, SARIMAX, Prophet, XGBoost, LSTM, TimeGPT**, enabling accurate traffic prediction, capacity planning, and anomaly detection across Ethernet & ILL backbones.
- Designed production-grade ML pipelines with dynamic hyperparameter tuning, feature engineering, **Boosting Ensembles** (Prophet + XGBoost), and a **3-layer Stacking Ensemble** (**XGBoost + LSTM + Meta-Model**), successfully deployed for real-time inference.
- Developed anomaly detection engines using **Z-Score, IQR, Isolation Forest, LOF, One-Class SVM** and clustering models including **K-Means** and **DBSCAN** for customer segmentation and traffic behavior analysis.

Analyst Intern, Accenture – Bangalore, Karnataka March 2022 – Jan 2023

- Developed **dynamic dashboards** and **data-driven UI** using **React.js**, **Tailwind CSS**, and **Redux-Saga**, enabling seamless visualization of campaign performance and engagement metrics.
- Utilized **Python**, **MongoDB**, and **SQL** for querying, preprocessing, and implementing **dynamic data filtering** powering interactive visual elements.
- Worked with **cross-functional teams** to define KPIs, optimize **API performance**, and design efficient **JSON payload structures** for streamlined communication.

Certifications

- Google IT Automation with Python Professional Certificate.
- Building Modern Python Applications on AWS.
- Designing and Implementing Azure AI Solutions.
- Google Cloud Professional Data Engineer.
- Red Hat.

Projects

VitaCount (MyChiffre) – AI Financial Operating System

- Built an **AI financial operating system** with **Next.js**, **TypeScript**, and **Supabase Postgres** + **RLS** multi-tenant isolation.
- Designed a **posting engine** that turns invoices, bills, and bank lines into GL entries, with a **balance-enforcing Postgres trigger** that rejects unbalanced journal entries.
- Delivered **AR/AP**, **Claude Vision receipt OCR**, **Plaid bank reconciliation**, and **FIFO inventory** accounting.
- Added an agentic layer with **reconciliation**, **AP bookkeeping**, and **AR collections agents** using tool-calling.

Shantiban City 3D Walkthrough

- Built a browser-based **3D walkthrough** of a real township in **three.js**, with all **220 plots traced from the master plan**.
- Rendered a **physically based sky** matched to the site's latitude, optimized to run smoothly on **mobile devices**.

AI Chatbot & n8n Automation Pipeline

- Built a chat-driven DevOps automation system using **n8n**, **OpenAI**, and **Terraform Cloud**, enabling users to provision AWS infrastructure through natural language commands.
- Designed AI workflows that convert user prompts into validated **Terraform templates** for EC2, S3, RDS, VPC, IAM, and Security Groups, ensuring fully automated, error-free provisioning.
- Integrated **remote Terraform workspaces** with dynamic variable injection, automated runs, and real-time status updates returned directly to the chatbot interface.
- Delivered an end-to-end **Conversational Infrastructure-as-Code (CIaC)** pipeline that eliminated manual configuration and demonstrated advanced cloud automation and AI orchestration capabilities.

Competitions And Awards

Extra Mile Award *Aug 2024 | Tata Communications*

- Received the "**Extra Mile Award**" for designing and developing an automation framework that dynamically generated **JSON configuration files** for diverse **BGP session setups**.
- Integrated the solution with Tata Communications' private **IZO Cloud platform**, enabling seamless deployment via **Ansible playbooks**.